Furuta pendulum with a stepper arm: model, linearisation and LQR design — worked calculation

2026-11-04

First hardware balance test passed (2026-11-04, one run)

Prepared for typesetting (LaTeX-ready). Every number below was recomputed independently (numpy, DARE solved by iteration) and matches `analysis/lqr_design.py`.

Evidence labels used throughout. [D] = derivation (pure mathematics). [I] = identified from our own measurements. [S] = simulation only. [H] = measured on hardware. [A] = assumption. **First hardware balance test passed (2026-11-04, one run).** Disturbance response, repeatability and swing-up not yet tested.

1 Coordinates, signs and units

Symbol	Meaning	Unit
θ	arm angle about the vertical motor axis; positive = clockwise seen from above (the `f` command) [H]	rad
α	pendulum angle about the hinge, measured from **hanging down**; positive = the AS5600 count direction	rad
$d = \alpha - \pi$	pendulum angle from **upright**; $d = 0$ is the balance point	rad
$L_r = 0.070$	motor axis to pendulum hinge (measured, approximate)	m
$m_p = 0.012$	pendulum mass (measured)	kg
l_p	hinge to pendulum centre of mass (**not yet measured**)	m
J_p	pendulum inertia about the hinge	kg m^2
b_p	equivalent viscous hinge damping	N m s
$g = 9.81$	gravity	m/s^2
$u = \ddot{\theta}$	model input: arm angular acceleration	rad/s^2

Stepper: 1/16 microstepping, 3200 steps per revolution, i.e. $3200/360 = 8.889$ steps per degree [H, verified: 800 steps = 90°].

Note: the instructor handout orders the state as $[\alpha, \dot{\alpha}, \theta, \dot{\theta}]$ and our code uses $x = [\theta, d, \dot{\theta}, \dot{d}]$. The ordering is a bookkeeping choice; the sign conventions above are what matter. [A: the handout's own sign convention for θ and α was not compared line by line.]

2 Why the model input is acceleration, not torque

General (torque) model. Let the arm have inertia J_r and viscous friction b_r about the motor axis, driven by motor torque τ. The Euler–Lagrange equations for the two coordinates $q = (\theta, \alpha)$

are

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} = Q_i, \qquad Q_\theta = \tau - b_r\dot{\theta}, \quad Q_\alpha = -b_p\dot{\alpha}.$$

This gives one **arm equation** (it contains τ, J_r, b_r) and one **pendulum equation** (it does not contain τ).

Reduction for a stepper. A stepper that does not miss steps imposes the arm motion $\theta(t)$ itself. Then θ is no longer an unknown: $\ddot{\theta} = u$ is a **prescribed function of time**. The pendulum equation contains θ only through $\dot{\theta}, \ddot{\theta}$, so it can be solved with u as input. The arm equation is no longer needed for the dynamics; it only tells us what torque the motor would have to deliver:

$$\tau_{\text{required}} = \big(J_r + m_p L_r^2 + \dots\big)u + \text{coupling terms} + b_r\dot{\theta}.$$

Therefore J_r, b_r and all motor constants **drop out of the design**. They only decide whether the motor can follow the commanded u without missing steps — a hardware limit that is **not verified** (see §10).

3 Pendulum equation (Lagrange) [D]

Take unit vectors $e_r = (\cos\theta, \sin\theta, 0)$, $e_\theta = (-\sin\theta, \cos\theta, 0)$, $e_z = (0, 0, 1)$ (z up). Let the pendulum swing in the plane spanned by e_θ and e_z, with the angle α measured from the downward vertical and positive along $+e_\theta$ (derivation convention). The centre of mass is

$$\mathbf{r} = L_r e_r + l_p \sin\alpha\, e_\theta - l_p \cos\alpha\, e_z.$$

Using $\dot{e}_r = \dot{\theta} e_\theta$, $\dot{e}_\theta = -\dot{\theta} e_r$:

$$\dot{\mathbf{r}} = (L_r\dot{\theta} + l_p\cos\alpha\,\dot{\alpha})\, e_\theta - l_p \sin\alpha\,\dot{\theta}\, e_r + l_p \sin\alpha\,\dot{\alpha}\, e_z,$$

$$|\dot{\mathbf{r}}|^2 = L_r^2\dot{\theta}^2 + 2L_r l_p \cos\alpha\,\dot{\theta}\dot{\alpha} + l_p^2\dot{\alpha}^2 + l_p^2 \sin^2\alpha\,\dot{\theta}^2.$$

Kinetic energy of the pendulum, for a slender pendulum (inertia about its own axis neglected) with $J_p = I_c + m_p l_p^2$:

$$T_p = \tfrac{1}{2} m_p L_r^2\dot{\theta}^2 + m_p L_r l_p \cos\alpha\,\dot{\theta}\dot{\alpha} + \tfrac{1}{2} J_p\dot{\alpha}^2 + \tfrac{1}{2} J_p \sin^2\alpha\,\dot{\theta}^2, \qquad V = -m_p g l_p \cos\alpha.$$

The α equation:

$$\frac{d}{dt}\frac{\partial T}{\partial \dot{\alpha}} = J_p\ddot{\alpha} + m_p L_r l_p\big(\cos\alpha\,\ddot{\theta} - \sin\alpha\,\dot{\theta}\dot{\alpha}\big),$$

$$\frac{\partial T}{\partial \alpha} = -m_p L_r l_p \sin\alpha\,\dot{\theta}\dot{\alpha} + J_p \sin\alpha\cos\alpha\,\dot{\theta}^2, \qquad \frac{\partial V}{\partial \alpha} = m_p g l_p \sin\alpha.$$

The $\dot{\theta}\dot{\alpha}$ terms cancel, leaving

$$\boxed{J_p\ddot{\alpha} + b_p\dot{\alpha} + m_p g l_p \sin\alpha + m_p L_r l_p \cos\alpha\,\ddot{\theta} - J_p \sin\alpha\cos\alpha\,\dot{\theta}^2 = 0.}$$

Sign convention. With α positive along $+e_\theta$ (the direction of arm motion) the coupling term has coefficient $+m_p L_r l_p \cos\alpha$ on the left, i.e. $s = -1$ in the form below. Physically correct: accelerating the hinge forward makes the bob lag *backwards*. If instead the encoder counts positive in the opposite direction, $\alpha \to -\alpha$ flips the sign and $s = +1$. We therefore write

$$J_p\ddot{\alpha} + b_p\dot{\alpha} + m_p g l_p \sin\alpha = s\, m_p l_p L_r \cos\alpha\,\ddot{\theta} + J_p \sin\alpha\cos\alpha\,\dot{\theta}^2, \qquad s = \pm 1.$$

s depends on the direction in which the AS5600 counts increase relative to the arm's forward direction. It is **found from the 1 Hz sine test** (§6), not assumed.

Centrifugal term. The term $J_p \sin\alpha \cos\alpha\, \dot\theta^2$ is dropped. Near upright, $\sin\alpha\cos\alpha = \sin d \cos d \approx d$, so the term is $\approx J_p d\, \dot\theta^2$ — a product of the small variable d and $\dot\theta^2$. Compared with the gravity restoring term it modifies the stiffness by $\Delta(\omega_n^2) \sim \dot\theta^2$, so it is negligible when

$$\dot\theta^2 \ll \omega_n^2 \approx 146 \text{ (rad/s)}^2.$$

At $\dot\theta = 1$ rad/s the ratio is $\dot\theta^2/\omega_n^2 = 0.7\%$. At the sketch's speed limit (2π rad/s) it is 27%, so the linear model is **not** valid at that speed. Balancing should stay at small speeds. [A, D]

4 Normalised form [D]

Divide by J_p and define

$$\omega_n^2 = \frac{m_p g l_p}{J_p}, \qquad 2\zeta\omega_n = \frac{b_p}{J_p}, \qquad l_{eq} = \frac{J_p}{m_p l_p} = \frac{g}{\omega_n^2}, \qquad K = \frac{L_r}{l_{eq}}.$$

Then, with $u = \ddot\theta$ and the centrifugal term neglected,

$$\boxed{\ddot\alpha + 2\zeta\omega_n\dot\alpha + \omega_n^2 \sin\alpha = s\, K \cos\alpha\; u.}$$

Only ω_n, ζ and K appear. m_p and l_p drop out of the design (they are needed only for the handout's parameter sheet).

Bound on l_p. Since $J_p = I_c + m_p l_p^2 \geq m_p l_p^2$, we have $l_{eq} = J_p/(m_p l_p) \geq l_p$, so $l_p \leq l_{eq}$. With the identified $l_{eq} = 6.7$ cm and a 7 cm pendulum, the mass must be concentrated near the tip ($l_{eq} = 7$ cm would be a point mass at the tip). A uniform rod of length 7 cm would give $l_{eq} = \frac{2}{3} \cdot 7 = 4.7$ cm, so it is not a uniform rod.

5 Identified numbers [I]

5.1 Natural frequency. The free-decay period (arm held, motor off) is $T = 0.516$ s (run 1) and 0.523 s (run 2); we use $T = 0.52$ s. The damping is light, so $\omega_d \approx \omega_n$:

$$\omega_n = \frac{2\pi}{T} = \frac{2\pi}{0.52} = 12.083 \text{ rad/s}, \qquad \omega_n^2 = 146.0 \text{ (rad/s)}^2.$$

5.2 Equivalent length.

$$l_{eq} = \frac{g}{\omega_n^2} = \frac{9.81}{146.0} = 0.06719 \text{ m} = 6.72 \text{ cm}.$$

5.3 Input gain.

$$K = \frac{L_r}{l_{eq}} = \frac{0.070}{0.06719} = 1.0418.$$

5.4 Damping ratio. The logarithmic decrement between successive same-side peaks is $\delta = \ln(x_n/x_{n+1})$. Run 1 peaks $-86.1^\circ \to -59.5^\circ$ and $+54.7^\circ \to +36.5^\circ$:

$$\delta = \ln\frac{86.1}{59.5} = 0.370, \qquad \delta = \ln\frac{54.7}{36.5} = 0.405.$$

Using $\zeta = \delta/\sqrt{4\pi^2 + \delta^2}$:

$$\zeta(0.370) = 0.059, \qquad \zeta(0.405) = 0.064, \qquad \zeta(0.46) = 0.073.$$

Later cycles give $\delta = 0.58, 0.69$ ($\zeta = 0.09, 0.11$): the decrement is **not constant**, which a viscous model would require. We use the first-cycle value

$$\zeta \approx 0.06 \quad \text{(equivalent viscous; range 0.05–0.07 on the first cycle).}$$

5.5 Dry (Coulomb) friction. The amplitude falls by an almost constant $\Delta \approx 11.4^\circ$ per half cycle. For a pendulum with restoring torque $\approx m_p g l_p \alpha$ (small-angle approximation) and constant friction torque T_c, the equilibrium shifts to $\pm T_c/(m_p g l_p)$ and the amplitude drops by $2T_c/(m_p g l_p)$ each half cycle:

$$\frac{T_c}{m_p g l_p} = \frac{\Delta}{2} = \frac{11.4^\circ \cdot \pi/180}{2} = \frac{0.199}{2} = 0.0995 \approx 0.10.$$

With $m_p = 0.012$: $T_c = 0.10 \cdot 0.012 \cdot 9.81 \cdot l_p = 1.18 \times 10^{-2}\, l_p$ N m (l_p in m). The formula is a small-angle approximation applied to swings up to 86°, so it is only indicative.

5.6 Inertia and damping (need l_p).

$$J_p = m_p l_p l_{eq} = 0.012 \cdot 0.06719 \cdot l_p = 8.06 \times 10^{-4}\, l_p \text{ kg m}^2, \qquad b_p = 2\zeta\omega_n J_p.$$

These stay symbolic until l_p is measured (balance the pendulum on an edge). They are **not needed for the LQR design**.

6 Sign s and model check against the open-loop sine tests [I]

For the hanging pendulum ($\alpha \approx 0$, $\sin\alpha \approx \alpha$, $\cos\alpha \approx 1$) driven by $\theta = \Theta e^{j\omega t}$, $\ddot{\theta} = -\omega^2\theta$:

$$\big(\omega_n^2 - \omega^2 + j\,2\zeta\omega_n\omega\big)\alpha = -s\,K\omega^2\theta \implies \frac{\alpha}{\theta}(j\omega) = \frac{-s\,K\omega^2}{\omega_n^2 - \omega^2 + j\,2\zeta\omega_n\omega}.$$

Magnitude:

$$\left|\frac{\alpha}{\theta}\right| = \frac{K\omega^2}{\sqrt{(\omega_n^2 - \omega^2)^2 + (2\zeta\omega_n\omega)^2}}.$$

Meaning of the sign. Well below resonance ($\omega \ll \omega_n$) the denominator is real and positive, so $\alpha/\theta = -s \cdot$ (positive): for $s = +1$ the pendulum is 180° out of phase with the arm; for $s = -1$ it is in phase.

1 Hz test ($\omega = 6.283$ rad/s): $K\omega^2 = 1.0418 \cdot 39.48 = 41.1$; denominator $146.0 - 39.5 = 106.5$ real, imaginary part $2 \cdot 0.06 \cdot 12.08 \cdot 6.283 = 9.1$, modulus 106.9:

$$\left|\frac{\alpha}{\theta}\right| = \frac{41.1}{106.9} = 0.385, \qquad \text{phase} = 180^\circ - \arctan\frac{9.1}{106.5} = 175^\circ \ (s = +1).$$

Measured [H]: ratio 0.31, phase $\approx 156^\circ$. The phase says the pendulum moves opposite to the arm, so **$s = +1$** [I]. The 19° phase difference is not explained by the model (encoder nonlinearity, arm angle is the commanded value, dry friction).

2 Hz test ($\omega = 12.566$ rad/s): $K\omega^2 = 1.0418 \cdot 157.9 = 164.5$; denominator real part $146.0 - 157.9 = -11.9$, imaginary part $2 \cdot 0.06 \cdot 12.083 \cdot 12.566 = 18.2$, modulus 21.8:

$$\left|\frac{\alpha}{\theta}\right| = \frac{164.5}{21.8} = 7.56, \qquad \text{phase} = 180^\circ - \big(180^\circ - \arctan\tfrac{18.2}{11.9}\big) = 57^\circ.$$

Measured [H]: ratio 4.9, phase 81° at a 40° amplitude.

Correction to an earlier statement. The predicted phase at 2 Hz is 57°, not $\approx 90^\circ$; 90° occurs only exactly at $\omega = \omega_n$. The measurement (81°) is closer to resonance than the model places it, which suggests the *effective* ω_n at that amplitude is closer to 12.6 rad/s than 12.08. Near resonance

both the ratio and the phase are very sensitive to ω_n and ζ: for example $\omega_n = 12.5$ rad/s (with K recomputed) gives a ratio of 9.3 and a phase of 85°, and $\omega_n = 12.3$ gives 8.7 and 70°.

Uncertainty of ω_n [I]. Zero-crossing full periods from the two logs are $0.531, 0.523, 0.517, 0.538$ s (run 1) and $0.556, 0.539, 0.528, 0.526, 0.515$ s (run 2), i.e. $\omega_n = 11.3$–12.2 rad/s, with the period longer at large amplitude. With 10 ms sampling and the ±5° encoder nonlinearity the uncertainty is about ±3%. Because $l_{eq} \propto 1/\omega_n^2$, that gives ±6% in l_{eq} and in K. **Re-measure ω_n at small amplitude ($\leq 20°$), three times**, before trusting K to better than 10 %. The simulation was run with $K \pm 20\%$ (§9).

Conclusion. The model captures the right order of magnitude and the sign. The comparison is not a validation; the over-prediction is consistent with dry friction, the 40° nonlinear amplitude and the encoder error. [I]

7 Linearisation about upright [D]

Set $\alpha = \pi + d$. Then $\sin\alpha = -\sin d \approx -d$, $\cos\alpha = -\cos d \approx -1$, and $\dot{\alpha} = \dot{d}$. Substituting into the normalised equation:

$$\ddot{d} + 2\zeta\omega_n\dot{d} - \omega_n^2 d = -s\,K\,u \implies \boxed{\ddot{d} = \omega_n^2 d - 2\zeta\omega_n\dot{d} - s\,K\,u.}$$

With $x = [\theta,\ d,\ \dot{\theta},\ \dot{d}]^\top$, $\ddot{\theta} = u$:

$$\dot{x} = Ax + Bu, \quad A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & \omega_n^2 & 0 & -2\zeta\omega_n \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 1 \\ -sK \end{bmatrix}.$$

Numerically ($s = +1$, $\omega_n^2 = 146.0$, $2\zeta\omega_n = 1.450$, $K = 1.0418$):

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 146.0 & 0 & -1.450 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1.0418 \end{bmatrix}.$$

The first two rows are kinematics ($\dot{\theta} = \dot{\theta}$, $\dot{d} = \dot{d}$). Row 3 says the arm acceleration is the input. Row 4 is the pendulum.

8 Stability and controllability [D, I]

Characteristic equation. The first two rows couple to rows 3–4 only through the lower-right structure; expanding $\det(sI - A)$:

$$\det(sI - A) = s^2\big(s^2 + 2\zeta\omega_n s - \omega_n^2\big) = 0.$$

Roots: $s = 0$ (double, the arm integrator chain) and

$$s = -\zeta\omega_n \pm \omega_n\sqrt{1 + \zeta^2}.$$

Numerically $\zeta\omega_n = 0.725$, $\omega_n\sqrt{1+\zeta^2} = 12.083 \cdot 1.0018 = 12.105$:

$$s_1 = -0.725 + 12.105 = +11.38, \qquad s_2 = -0.725 - 12.105 = -12.83 \text{ rad/s}.$$

One positive real root: the upright is unstable, with a time constant $1/11.38 = 88$ ms. That is why the loop must run much faster than this (200 Hz = 5 ms is ≈ 18 samples per time constant).

Controllability matrix. $C = [B \;\; AB \;\; A^2B \;\; A^3B]$ with

$$AB = \begin{bmatrix} 1 \\ -K \\ 0 \\ 2\zeta\omega_n K \end{bmatrix}, \quad A^2B = \begin{bmatrix} 0 \\ 2\zeta\omega_n K \\ 0 \\ -K\omega_n^2(1+4\zeta^2) \end{bmatrix}, \quad A^3B = \begin{bmatrix} 0 \\ -K\omega_n^2(1+4\zeta^2) \\ 0 \\ 2\zeta\omega_n^3 K(2+4\zeta^2) \end{bmatrix}.$$

Numerically (columns B, AB, A^2B, A^3B; rows $\theta, d, \dot\theta, \dot d$):

$$C = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -1.042 & 1.511 & -154.29 \\ 1 & 0 & 0 & 0 \\ -1.042 & 1.511 & -154.29 & 444.26 \end{bmatrix}.$$

Expand the determinant along row 1 (its only nonzero entry is the 1 in column 2, cofactor sign -1), then along the row of the remaining minor that contains a single 1 (cofactor sign -1). The two sign flips cancel, leaving $\det C = \Delta$ with Δ the 2×2 determinant of rows $d, \dot d$ and columns A^2B, A^3B:

$$\Delta = \begin{vmatrix} 2\zeta\omega_n K & -K\omega_n^2(1+4\zeta^2) \\ -K\omega_n^2(1+4\zeta^2) & 2\zeta\omega_n^3 K(2+4\zeta^2) \end{vmatrix} = K^2\omega_n^4\Big[4\zeta^2(2+4\zeta^2) - (1+4\zeta^2)^2\Big].$$

The bracket simplifies to $8\zeta^2 + 16\zeta^4 - 1 - 8\zeta^2 - 16\zeta^4 = -1$, so $\Delta = -K^2\omega_n^4$ and

$$\boxed{\det C = -K^2\omega_n^4 = -(1.0418)^2(146.0)^2 = -2.31\times 10^4 \neq 0.}$$

The rank is 4 for every ζ, as long as $K \neq 0$ and $\omega_n \neq 0$. The system is controllable [D]; the numerical rank computed by the script is also 4.

9 Discrete LQR design [D, S]

Cost. Minimise $J = \sum_{k=0}^{\infty} \big(x_k^\top Q x_k + u_k^\top R u_k\big)$ subject to $x_{k+1} = A_d x_k + B_d u_k$.

Discretisation (zero-order hold, $\Delta t = 5$ ms).

$$\begin{bmatrix} A_d & B_d \\ 0 & 1 \end{bmatrix} = \exp\!\left(\begin{bmatrix} A & B \\ 0 & 0 \end{bmatrix}\Delta t\right).$$

Weights (Bryson's rule: $1/(\text{acceptable value})^2$).

$$Q = \operatorname{diag}\big(\tfrac{1}{0.5^2}, \tfrac{1}{0.08^2}, \tfrac{1}{3^2}, \tfrac{1}{5^2}\big) = \operatorname{diag}(4,\ 156.25,\ 0.1111,\ 0.04), \qquad R = \tfrac{1}{15^2} = 4.444\times 10^{-3},$$

that is 0.5 rad of arm drift, 0.08 rad (4.6°) pendulum error, 3 rad/s arm speed, 5 rad/s pendulum speed, 15 rad/s^2 acceleration. These are design choices [A]: tighter weights balanced in simulation but chattered because of encoder quantisation and derivative noise.

Method (Riccati). Solve the discrete algebraic Riccati equation

$$P = A_d^\top P A_d - A_d^\top P B_d\,(R + B_d^\top P B_d)^{-1} B_d^\top P A_d + Q,$$

and form the gain

$$K_d = (R + B_d^\top P B_d)^{-1} B_d^\top P A_d, \qquad u_k = -K_d x_k.$$

It was solved by fixed-point iteration (and by `scipy.linalg.solve_discrete_are` in the script). Both give

$$\boxed{K_d = \big[-27.555,\ -600.497,\ -14.217,\ -44.909\big], \qquad u = -\big(k_1\theta + k_2 d + k_3\dot\theta + k_4\dot d\big).}$$

The closed-loop discrete eigenvalues have moduli 0.9308 (pair) and 0.9833 (pair), all < 1; as continuous poles $\ln\lambda/\Delta t$: $-14.3 \pm 6.9j$ and $-3.4 \pm 2.4j$ rad/s. The fast pair is faster than the open-loop unstable pole (11.4 rad/s), as it must be.

Sanity check 1 — signs. $k_2 < 0$ means $u = -k_2 d = +600\,d$: for $d > 0$ the arm accelerates in $+\theta$. From $\ddot{d} = \omega_n^2 d - sKu$ with $s = +1$ this makes $\ddot{d}$ more negative and pushes d back to zero — the right correction direction. If s were -1 the same gains would destabilise the loop, which is why s must be right.

Sanity check 2 — pendulum alone. Keep only k_2, k_4 (ignore the arm states). Then

$$\ddot{d} = (\omega_n^2 + Kk_2)\,d + (-2\zeta\omega_n + Kk_4)\,\dot{d} = -479.6\,d - 48.24\,\dot{d},$$

with roots -14.0 and -34.2 rad/s. Stability requires $\omega_n^2 + Kk_2 < 0$, i.e. $|k_2| > \omega_n^2/K = 146.0/1.0418 = 140.1$; our 600.5 is 4.3 times that limit. The -14.0 rad/s root matches the fast closed-loop pole above.

Sanity check 3 — static (steady-state) condition. In steady state $u = 0$, $\ddot{\theta} = 0$, $\dot{\theta} = 0$, $\dot{d} = 0$. Row 4 then forces $d = 0$. If the pendulum angle used by the controller has a calibration bias b ($d_{meas} = d + b$), $u = 0$ requires $k_1\theta + k_2 b = 0$:

$$\theta_{ss} = -\frac{k_2}{k_1}\,b = -\frac{-600.497}{-27.555}\,b = -21.8\,b.$$

A 1° bias therefore makes the arm sit $\approx 22°$ off centre, matching the simulation ($-21.7°$). The sketch's slow auto-trim $\dot{c} = -k_i\theta$ ($k_i = 0.023$) integrates this offset away; it is an addition to the LQR, outside the LQR optimality proof [S].

Simulation results [S, not hardware]. A nonlinear simulation (5° start, quantised encoder, filtered derivative, 1-sample delay, $|u| \le 100$ rad/s^2) balances for the nominal plant and with $K \times 0.8$ and $K \times 1.2$; with a 1° bias the trim removes the 22° arm offset. It does **not** include dry hinge friction, the ±5° encoder nonlinearity or motor torque limits.

10 Implementation notes and what they assume

- **Speed command.** The controller integrates u to a speed command, $\dot{\theta} \leftarrow \dot{\theta} + u\,\Delta t$, and uses the commanded speed as $\dot{\theta}$ in the feedback. This assumes the stepper really follows the commanded speed.
- **Derivative.** $\dot{d}$ comes from a difference of the encoder angle through a 25 Hz low-pass ($\tau = 1/(2\pi \cdot 25) = 6.4$ ms). The filter delay is not in the LQR design.
- **Untested motor limits.** The arm test (§ motion_core) only reached 60°/s (533 steps/s). The sketch allows 2π rad/s = 3200 steps/s and $|u| \le 100$ rad/s^2 $= 5.1 \times 10^4$ steps/s^2. A stepper at 12 V loses torque at high step rates, so **missed steps at these limits are likely and undetectable** (no arm encoder).

11 Assumptions and limitations (summary)

1. **No arm encoder.** $\theta, \dot{\theta}$ are the *commanded* values. Missed steps cannot be detected; the 90° test only verifies small slow moves. [H]
2. **Dry hinge friction.** About 10% of the maximum gravity torque; the pendulum stops abruptly and rests off-centre. The viscous model with $\zeta \approx 0.06$ is only an equivalent. Dry friction adds a dead band near upright.

3. **Encoder nonlinearity** about $\pm 5^\circ$ (weak magnet). It limits the accuracy of ω_n, ζ and the zero, and it is the same order as the controller's workspace ($\pm 4^\circ$ catch window).

4. **Identified parameters are uncertain:** $\omega_n \pm 3\%$, hence $K \pm 6\%$; L_r is approximate; l_p, J_p, b_p are not measured.

5. **Sign conventions.** $s = +1$ was read from the 1 Hz phase (156° vs predicted 175°). The user has **not yet chosen** the final direction convention for θ; if the encoder direction or the arm direction is changed, s must be re-derived.

6. **Linear model validity.** Needs small d and $\dot{\theta}^2 \ll \omega_n^2$ (§3). No swing-up is covered here.

7. **Delay and filtering** are not in the LQR design (only in the simulation).

8. **Hardware balancing:** tested in one run; disturbance response, repeatability and swing-up not yet tested. The motor's ability to deliver u across its full limits and the gains' robustness to friction remain unverified beyond this run.

11.1 Robustness to hinge dry friction (simulation)

A crude simulation with stiction plus Coulomb friction of $T_c = 0.10 m_p g l_p$ (the value estimated from the decay tests) predicts that the designed gains do not settle. The pendulum sticks about 0.9° off upright and the arm runs away to the $\pm 90^\circ$ software limit.

Tighter LQR gains ($K = [-68.2, -1660.9, -40.1, -94.7]$) give a limit cycle of about $\pm 9^\circ$ on the pendulum and about $\pm 105^\circ$ on the arm.

With $T_c = 0.05 m_p g l_p$ the result is still a limit cycle of about $\pm 5^\circ$ on the pendulum and about $\pm 50^\circ$ on the arm. Without friction it settles to within $\pm 0.07^\circ$.

The simulation is indicative only (crude friction model): see `analysis/friction_check.py` and `analysis/friction_check_output.txt`.

Recommended next step: reduce the hinge friction mechanically, and check it with a driver-off "does it stand tilted?" test.

On hardware the controller balanced despite the estimated friction, so this crude friction simulation was too pessimistic; the slow ± 10–25° arm wander may partly come from friction and encoder error.

12 First hardware balance test

The first hardware balance test on 2026-11-04 balanced for 10.0 s and was still balancing when the log ended. [H]

Quantity	Measured result
Pendulum	max 3.6° from upright; rms 0.87° after 2 s
Arm	wandered between -9.9° and $+24.9^\circ$
Control u	rms 18.5 rad/s^2; at the limit 0.3% of the time
Auto-trim	converged to -1.10°
Run limits	100 rad/s^2 and 360°/s (earlier limits)

Source: `docs/BALANCE_TEST_2026-11-04.md`; measured log `data/balance_2026-11-04_1427.csv`.

13 Next steps (evidence required)

1. Re-measure ω_n at $\leq 20^\circ$, three runs; recompute l_{eq}, K.
2. Measure l_p (edge balance) to complete J_p, b_p, T_c.
3. Choose and record the θ direction convention; recheck s.
4. Hardware test of the balance controller with bounded limits (small V_{max}, U_{max}) and the emergency stop; log θ, d, u, trim at 100 Hz; compare with the simulation.